REVISION TEST -III

CLASS : 12TH STD. **MARKS : 70**

SUBJECT : CHEMISTRY **TIME : 2.30HRS**

PART-I

Chosse the correct answers. **15 X 1 =15**

1. Which of the following can act as Lowry – Bronsted acid as well as base?

a) HCl b) SO_4^{2-} c) HPO_4^{2-} d) Br^-

2. Which of the following electrolytic solution has the least specific conductance

a) 2N b) 0.002N c) 0.02N d) 0.2N

3. Fog is colloidal solution of

a) solid in gas. b) gas in gas c) liquid in gas d) gas in liquid

4. When phenol is distilled with zinc dust it gives

a)Benzaldehyde b)benzoic acid c)benzene d) tolune

5. The compound that does not undergoes cannizzaro reaction is

a)Formaldehyde b)benzaldehyde

c)acetaldehyde d)trimethyl acetaldehyde

6. The product formed by the reaction an aldehyde with a primary amine

a) carboxylic acid b) aromatic acid

c) schiff 's base d) ketone

7. The pyrimidine bases present in DNA are

a) Cytosine and Adenine b) Cytosine and Guanine

c) Cytosine and Thiamine d) Cytosine and Uracil

8. Which of the following is used for concentrating ore in metallurgy?

a) Leaching b) Roasting c) Froth floatation d) Both (a) and (c)

9. The metalloid among the following is

a) Pb b) P c) Ge d) Sn

10. In which of the following , NH_3 is not used?

a) Nessler's reagent b) Reagent for the analysis of IV group basic radical

c) Reagent for the analysis of III group basic radical d) Tollen's reagent

11. The magnetic moment of Mn^{2+} ion is

a) 5.92BM b) 2.80BM c) 8.95BM d) 3.90BM

12 which of the following is paramagnetic in nature

a) $[Zn(NH_3)^4]^{2+}$ b) $[Co(NH_3)_6]^{3+}$

c) $[Ni(H_2O)_6]^{2+}$ d) $[Ni(CN)_4]^{2-}$

13. The total number of atoms per unit cell in fcc

a)1 b)2 c)3 d)4

14. The rate constant of a reaction is $5.8 \times 10^{-2}\ s^{-1}$ The order of the reaction is
    a) First order b) zero order c) Second order d) Third order
15. Which one of the following is ziegler natta catalyst
    a) $CO_2(CO)_8$ b) Rh/Ir complex
    c) $TiCl_4 + Al(C_2H_5)_3$ d) none of these

PART-II

Answer the following any six questions. 6 X 2 = 12

Question number 24 is compulsory.

16. what is inert pair effect ?
17. Which is more stable among Fe^{3+} and Fe^{2+} ? why ?
18. Define rate law and rate constant
19. How are point defect classified ?
20. Write the ostwald dilution law?
21. Name the factors affecting adoption
22. Write the reaction of benzaldehyde with chlorine in the absence and presence of catalyst.
23. How to differentiate phenol and alcohol.?
24. An aromatic simplest nitro compound A on reduction using Sn/HCl gives B. B undergoes carbylamine reaction . Identify A and B.

PART-III

Answer the following any six questions. 6 X 3 = 18

Question number 33 is compulsory.

25. Why do transition elements and its compound act as catalyst?
26. write the postulates of werner's theory
27. What is the action of heat on boric acid ?
28. Mention any two factors that affect electrolytic conductance.
29. Write the difference between SOl and gel
30. Write two test for aldehyde ?
31. Draw the structure of lactose and Sucrose
32. What is mean by electro osmosis?
33. Write the expression for the solubility product of $Ca_3(PO_4)_2$ and Hg_2Cl_2

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PART-IV

Answer the following all questions. **5 X 5 = 25**

34.a) I) write the difference between mineral and ore ?
ii)Explain about cynide leaching

(OR)

b) I) Difference between diamond and graphite
ii) what is catenation ? describe the catenation property of carbon

35 a) I) What is the reaction of ammonia with iron and copper salts ?
ii) What is meant by transitionelements ? give example

(OR)

b) I) Write the formula for the following coordination compounds
I) potassiumhexacyanidoferrate(II) ii) tetracarbonyl nickel(0)
ii) calculate the percentage efficiency of packing in case of body centered cubic crycrystal

36 a) I)the rate constant for a first order reaction is $1.54 \times 10^{-3}\ s^{-1}$. calculate its half life time
ii) Give three examples for zero order reaction

(OR)

b)I) Define ionic product product of water give its value at room temperature
ii) What are the conversion used Galvanic cell notation ?

37 a)I) Explain intermediate compound formation theory of catalysis with an example
ii) What is epimerization ?

(OR)

b) I) complete the following reaction

i) $CH_3 - CH_2 - OH \xrightarrow{P\,Br_3} A \xrightarrow{aq.NaOH} B \xrightarrow{Na} C$

ii) $C_6H_5 - OH \xrightarrow{Zndert} A \xrightarrow[Anhydrs\ AlCl_3]{CH_3Cl} B \xrightarrow{acid\ KMnO_4} C$

ii) Identify A and B $CH_3\text{-}CH{=}CH_2 + H_2O/H^+ \rightarrow A$

$CH_3\text{-}CH{=}CH_2 + B_2H_6 + H_2O_2/OH^- \rightarrow B$

38 a) Explain the structure of fructose.

(OR)

b)I)How will you distinguish between
primary secondary and tertiary amine?
ii) explain the diazitisation with suitable example

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